AI PLAYBOOK FOR PUBLIC HEALTH

AI Playbook and Toolkit for Public Health Departments

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Generative and Agentic AI

Implementation Playbook

for Public Health Departments

A Comprehensive Guide for State, Territorial, Local, and Tribal Health Departments

Executive Summary

Generative and agentic artificial intelligence (AI) are moving rapidly from research laboratories into the operational workflows of public health departments, offering measurable benefits in disease surveillance, health communication, epidemic modeling, health equity analysis, and administrative functions. This playbook provides state, territorial, local, and tribal (STLT) health departments with a practical, step-by-step framework for understanding, planning, implementing, and governing generative and agentic AI technologies.

Unlike a technical manual, this playbook is structured as a series of "plays" — sequenced, actionable steps that leaders and teams can follow. Each play explains what to do, who should be involved, and how to use the included tools and checklists to move from concept to sustainable practice. Checklists are support tools; the plays themselves describe how to lead the work.

This playbook includes a structured 12-Play AI Implementation Framework organized across four phases: PLAN, BUILD, DEPLOY, and GOVERN. The planning plays guide agencies through vision setting, stakeholder engagement, readiness assessment, workforce development, prioritization of use cases, and change management. The build plays cover roadmap design and funding strategy. The deploy plays address governance design and solution development. The govern plays focus on monitoring system performance and ensuring ongoing compliance with responsible AI policies.

The companion document — AI Playbook for Public Health: Tools and Checklists — contains 34 practical, fillable tools organized in the sequence in which they are used across the 12 plays. Use the playbook and the toolkit together.

Complete Crosswalk: Plays and Supporting Tools

This table shows each play, its primary output, and all supporting toolkit tools. Use it during each play to confirm you have the right tools ready.

Play

Play Name

Primary Output

Supporting Tools

Play 1

Vision & Guardrails

AI vision and principles document

Tools: 1, 3, 12, 13

Play 2

Establish AI Governance

Governance committee, policy, and charter

Tools: 3, 13, 15, 22, 23, 24

Play 3

Engage Stakeholders

Stakeholder map and engagement plan

Tools: 11, 14, 15

Play 4

Readiness Assessment

Readiness scores and gap register

Tools: 1, 2, 9

Play 5

Workforce Development

AI workforce training plan

Tools: 5, 16, 17, 18

Play 6

Prioritize Use Cases

Governance-approved AI use case portfolio

Tools: 2, 3, 11, 19

Play 7

Design the Roadmap

Governance-approved AI roadmap

Tools: 2, 4, 9, 10

Play 8

Develop a Funding Strategy

Funding portfolio and grant alignment

Tools: 4, 8, 10

Play 9

Change Management

Adoption strategy and communications plan

Tools: 5, 20, 21

Play 10

Build and Deploy AI Solutions

AI pilot and production deployments

Tools: 2, 4, 7, 9, 10, 23, 24, 25, 26, 27, 28

Play 11

Monitor and Evaluate AI Systems

Performance and equity monitoring reports

Tools: 6, 29, 30, 31

Play 12

Execute Governance and Oversight

Audits, compliance records, governance docs

Tools: 6, 7, 31, 32, 33, 34

1. Understanding Artificial Intelligence

What Are Generative and Agentic AI?

Artificial intelligence is not a single technology. It is a family of methods that enable computer systems to learn from data, recognize patterns, generate new content, and take action toward defined goals. For public health agencies, two branches of AI have emerged as especially consequential: generative AI and agentic AI. Understanding what these technologies are, how they differ from earlier AI approaches, and what they can realistically accomplish is the essential starting point for any agency considering adoption.

Generative AI

Generative AI refers to systems trained on large bodies of existing data that can produce new content — including text, images, code, and synthetic datasets — when prompted by a user. The most widely deployed generative AI systems are large language models (LLMs), which learn statistical patterns across billions of words of text and can generate fluent, coherent, and contextually appropriate responses to natural language queries.

The leading LLMs in current use include GPT-4 and GPT-4o from OpenAI (available via Microsoft Azure OpenAI), Claude from Anthropic, Gemini from Google DeepMind, and Llama from Meta (open-source). For most STLT agencies, the practical choice is between commercially hosted APIs with Business Associate Agreement (BAA) support — such as Azure OpenAI and Google Vertex AI — and open-source models that can be deployed within agency-controlled infrastructure.

For public health, the most immediately useful generative AI capabilities include:

Content generation: drafting health education materials, vaccination messages, press releases, and program protocols from natural language prompts.

Document retrieval and synthesis: answering questions about regulatory guidance, policy documents, or scientific literature by retrieving relevant passages and synthesizing a response — a technique known as retrieval-augmented generation (RAG).

Data interpretation: translating statistical outputs, epidemiological findings, or surveillance reports into plain-language summaries or infographics accessible to non-technical audiences.

Code generation: writing, debugging, and explaining programming code for data analysis, modeling, and visualization tasks.

Synthetic data generation: creating realistic but de-identified datasets that mimic the statistical properties of real health data, enabling model training and testing without exposing protected health information.

Important Guardrail

Generative AI should always be used with human review, clear guardrails, and strong privacy protections. It can dramatically reduce the time to produce communication and documentation — peer-reviewed studies document a 72–80% reduction in surveillance chart review time — but it must not replace expert judgment or local context.

Agentic AI

Agentic AI systems can not only analyze data or generate content, but also take a sequence of actions toward a defined goal across multiple systems, following explicit rules and with human oversight. For public health, the most immediately useful agentic AI capabilities include:

Coordinating multi-step workflows, such as pulling data from several systems, summarizing key findings, and routing a draft brief to the appropriate epidemiologist for review.

Monitoring incoming information streams (for example, news feeds or case reports), classifying and prioritizing items, and automatically creating task queues for surveillance or response teams.

Automating routine operational tasks, such as drafting follow-up messages to patients or partners and queuing them for staff approval and sending.

Orchestrating interactions between existing tools — dashboards, case management systems, communication platforms — so that when predefined thresholds are crossed, the right people receive the right information and suggested next steps, while retaining final decision authority.

Deep Research: A Practical Agentic AI Capability

One of the most immediately accessible forms of agentic AI for STLT health departments is "Deep Research," a capability now offered by several commercial AI vendors (including Google, OpenAI, and Anthropic). Unlike a standard chatbot that answers a single question, Deep Research tools autonomously plan and execute multi-step research tasks across the web, analyze and cross-reference findings from multiple sources in real time, and generate citation-based reports with transparent reasoning.

Based on CDC's internal exploratory evaluation, Deep Research tools are particularly well-suited for literature synthesis on emerging health threats or intervention strategies; policy and legal scans across jurisdictions; strategic planning and scenario analysis; and comparative analysis of programs, interventions, or regulations. These are tasks that currently consume significant staff time and are well-suited to AI-assisted first drafts that experts then validate and refine.

Limitations and Risks

Generative and agentic AI systems carry well-documented limitations that are especially consequential in public health settings. Every deployment decision should account for these risks.

Risk

Description

Mitigation

Hallucination

LLMs can generate plausible-sounding but factually incorrect outputs. In peer-reviewed studies, GPT-4 Turbo produced incorrect information in 13.4% of regulatory queries.

Require human adjudication of all consequential outputs before action.

Performance degradation with complexity

LLM accuracy drops substantially as task complexity increases, from 88.4% on binary yes/no questions to 74.5% on spatial inference tasks.

Match AI tools to task complexity; do not assume general-purpose LLMs perform well on all tasks.

Data and representational bias

Models trained on non-representative data will systematically underperform for underrepresented populations, especially tribal and rural communities.

Assess training data provenance; monitor subgroup performance continuously.

Privacy and data sovereignty

Sending protected health information to commercial LLM APIs without a BAA violates HIPAA and may violate tribal data sovereignty principles.

Establish appropriate legal agreements before any use of AI with identifiable or protected health information.

Automation bias

Staff who work alongside AI systems over time may reduce their critical scrutiny of AI outputs.

Governance protocols must actively counteract automation bias through regular audits and mandatory human review checkpoints.

2. Generative and Agentic AI in Public Health

Generative and agentic AI represent the most consequential shift in artificial intelligence for public health since the field began. Building on decades of computational methods, these technologies have emerged in the 2020s as transformative tools that STLT health departments can deploy today.

Generative AI in Practice

Departments are using large language models to draft plain-language health education materials in multiple reading levels and languages, which staff then review and adapt for local context. Generative tools help create first drafts of protocols, grant narratives, and policy documents, freeing staff time for higher-value work while retaining human review for accuracy and tone. Some agencies are exploring synthetic data generation — using generative models to create realistic but privacy-preserving datasets that can be used for training and testing AI systems without exposing identifiable health information.

Agentic AI in Practice

Public health and organizations in the United States have begun experimenting with agentic AI — systems that not only analyze data but can take multi-step actions across workflows under defined guardrails and human oversight. In outbreak response, for example, CDC and partners are testing AI systems that automatically ingest large volumes of open-source information each day, classify items by disease, location, and credibility, summarize key developments, and route high-priority events to epidemiologists, dramatically reducing manual scanning time while keeping humans in the loop for final judgment.

Key Lesson from History

Generative and agentic AI are most successful when built on strong data systems, aligned with clear public health objectives, and governed with attention to equity and trust. The strongest demonstrated outcomes are in surveillance efficiency, messaging effectiveness, and epidemic forecasting accuracy.

How AI Supports Public Health Workflows

Many public health activities follow a common workflow that moves from signal detection to decision support. Understanding this workflow helps clarify where generative and agentic AI tools may provide the greatest value.

Workflow Stage

Description

AI Applications

Signal Detection

Identifying signals that may indicate emerging health events or trends from surveillance systems, laboratory reporting, and environmental monitoring.

Anomaly detection algorithms; LLM-assisted monitoring of open-source information streams.

Evidence Gathering

Collecting relevant evidence to determine whether a signal represents a meaningful public health concern.

Agentic AI systems that retrieve datasets, compile scientific literature, and assemble supporting documentation.

Analysis

Evaluating collected information to determine significance.

AI tools supporting statistical analysis, exploratory data visualization, and pattern identification.

Synthesis

Synthesizing information from multiple sources into a coherent picture.

Generative AI tools that summarize research findings, consolidate reports, and generate draft evidence summaries.

Briefing and Communication

Communicating findings through dashboards, situation reports, policy briefs, or public health advisories.

Generative AI tools that draft reports, summarize trends, and adapt communications for different audiences.

Decision Support

Using available evidence to guide response activities and policy decisions.

AI tools that organize relevant information and generate structured briefing materials for leadership review.

3. Federal Funding Opportunities for AI

Public health departments do not need to fund AI initiatives from scratch. Several major federal funding streams can be braided together to support AI infrastructure, workforce, pilots, and sustainment. Understanding which programs apply — and how to position AI within them — is essential to building a sustainable funding strategy.

CDC Data Modernization Initiative (DMI)

The CDC Data Modernization Initiative is a multi-year effort to modernize core public health data systems, including surveillance, laboratory reporting, and data exchange. DMI funds are typically distributed through cooperative agreements to states, territories, large local jurisdictions, and tribal partners, with a focus on building interoperable, cloud-ready, standards-based infrastructure.

For AI, DMI is the backbone rather than a separate line item. Data platforms, APIs, and governance structures funded by DMI are exactly what AI projects need to succeed: high-quality, timely data that can be used safely and consistently across programs.

How to position AI within DMI:

Treat AI as a "capability layer" on top of DMI, not a competing project.

In your DMI work plans, explicitly describe how modernized data systems will support AI-enabled surveillance, forecasting, and decision support.

Use DMI resources to fund shared data infrastructure (cloud data lakes, data integration pipelines, standard vocabularies) that can serve multiple AI use cases.

Leverage DMI's emphasis on workforce to support data engineering, informatics, and analytics positions that will also staff AI initiatives.

Public Health Infrastructure Grant (PHIG)

The Public Health Infrastructure Grant provides broad, flexible funding to strengthen foundational public health capabilities over several years. PHIG explicitly supports workforce, data systems, organizational capacity, and health equity — all critical prerequisites for responsible AI.

Strategic use of PHIG for AI:

Workforce: Justify AI-related roles (data scientists, informaticians, AI project managers) as necessary to modern public health practice, not as "experimental" positions.

Data systems: Fund AI-ready analytics environments, visualization tools, and integration with surveillance and EHR systems.

Organizational capacity: Stand up an AI governance committee, innovation lab, or center of excellence that can coordinate projects and set standards across programs.

Health equity: Support community engagement, equity impact assessment, and ongoing monitoring of AI systems to ensure they reduce, rather than widen, disparities.

Funding Tip

PHIG allows you to build multi-year roadmaps. Use that flexibility to move from pilots to sustainable services, rather than one-off projects that end when the grant does.

Other Federal Funding Streams

Several other federal programs can be braided together to support AI work:

Funding Source

AI Applications

HRSA digital health and telehealth programs

Generative AI-assisted patient communication drafting, LLM-powered document search for regulatory guidance, and health equity analysis tools.

NIH / NSF smart health programs

Research funding and academic partnerships for generative AI applications, LLM fine-tuning for public health, and agentic AI system development.

SAMHSA technology-based services

Generative AI for early identification and intervention in opioid overdose prevention; LLM-assisted client communication and care coordination.

EPA environmental justice grants

GeoAI and LLM-powered environmental health analysis; heat risk mapping for historically underserved neighborhoods.

FEMA emergency management grants

Generative AI for disaster resilience in tribal and indigenous communities; agentic AI for multi-step outbreak forecasting workflows.

4. AI Maturity Model

Public health agencies vary widely in their readiness to adopt generative and agentic AI technologies. This AI Maturity Model helps agencies assess their current level of capability across five dimensions: leadership and culture, data infrastructure, workforce capacity, technology systems, and governance. It identifies the activities and milestones associated with each level and maps each level to the relevant plays in the AI Implementation Framework.

The model is not a grading system. A Level 2 agency that uses its readiness assessment to make a well-governed commitment to a first pilot is doing more responsible AI work than a Level 4 agency that has deployed tools without adequate equity oversight. Honesty about current state is more valuable than aspiring to a higher level.

Model Overview

Level	Stage	Organizational Focus	Primary Activities	Relevant Plays
1	Awareness	Learning and exploration	Exploring concepts, interim policy, informal experimentation, identifying AI lead	Plays 1, 2
2	Readiness	Planning and preparation	Readiness assessment, workforce planning, use case screening, roadmap and funding strategy	Plays 3, 4, 5, 7, 8
3	Pilots	Governance and initial development	Governance establishment, equity assessment, pilot development, model validation	Plays 6, 9, 10
4	Deployment	Operational integration	Full deployment, scaling, performance monitoring, equity auditing, sustainment planning	Plays 10, 11
5	Optimization			

Continuous improvement

Model refinement, capability expansion, peer contribution, sustained governance

Plays 11, 12

Level 1 | Awareness and Exploration

Building foundational understanding of generative and agentic AI in public health

Overview

At Level 1, the agency is in the earliest stages of learning about generative and agentic AI. There is no formal AI strategy, no dedicated governance structure, and no active pilot projects. Staff may be experimenting informally with publicly available AI tools, but these efforts are individual rather than organizational and occur without defined policies, oversight, or data protections.

This stage requires deliberate navigation. Informal, ungoverned AI use carries real risks: staff may inadvertently submit protected health information to external AI systems, rely on hallucinated outputs, or develop inconsistent practices that create organizational liability. The goal at Level 1 is not to delay action but to ensure that early exploration happens within a framework that protects the agency and the communities it serves.

Organizational Profile at This Level

Leadership & Culture

Data Infrastructure

Workforce Capacity

Technology Systems

Governance & Legal

Little or no executive awareness of AI implications for public health. AI is not part of strategic planning discussions.

No documented data readiness assessment. Data systems have not been evaluated for AI compatibility, API availability, or quality requirements.

Scattered individual interest in AI among staff. No formal AI literacy training. No staff with data science, informatics, or AI-specific roles.

No dedicated AI tools or environments. Staff access publicly available AI tools without IT oversight or data protection review.

No AI policies, governance structures, or responsible use guidelines. No legal review of AI-related risks has occurred.

Typical Activities

Leadership and key staff attend webinars, conferences, or peer agency briefings on generative and agentic AI in public health.

Internal discussions begin about potential AI use cases relevant to the agency's programs, such as health communications drafting or surveillance summarization.

Staff experiment informally with publicly available AI tools for low-stakes tasks such as drafting emails or summarizing meeting notes.

The agency identifies peer agencies and national partners (CDC, CSTE, ASTHO, NACCHO) with experience in public health AI adoption.

Leadership initiates discussion about risks, ethical considerations, and what responsible AI use would require from the organization.

An interim acceptable-use policy for AI tools is drafted or adopted to govern informal experimentation while a formal governance structure is developed.

Initial exploration of federal funding opportunities, such as PHIG and DMI, that could support AI infrastructure or workforce development.

Common Challenges at This Level

Staff enthusiasm outpacing organizational readiness, leading to ungoverned tool use with real privacy and accuracy risks.

Confusion between different types of AI: generative AI, predictive analytics, and traditional data tools are often conflated, creating unrealistic expectations.

Leadership unsure whether AI is relevant to their specific programs or scale of operation, leading to indefinite deferral.

No clear owner for AI exploration, so questions about responsible use go unanswered because no one has authority to make decisions.

Plays and Milestones

Relevant Plays

Key Milestones for Advancement

• Play 1: Vision & Guardrails

• Play 2: Engage Stakeholders

• At least one leadership discussion on AI opportunities and risks has occurred and been documented.

• An initial list of 3 to 5 potential AI use cases has been identified by program staff.

• Leadership has committed to completing a formal AI Readiness Self-Assessment (Tool 1).

• An interim acceptable-use policy for AI tools has been drafted or adopted.

• A lead staff member or team has been designated to coordinate early AI exploration.

Warning Sign: Stuck at This Level

The agency is stuck at Level 1 if informal AI experimentation is growing without any organizational response. Signs include staff using AI tools without policies in place, leadership deferring AI discussions indefinitely, or the agency completing AI-related trainings without moving toward a readiness assessment or governance commitment.

Goal for Advancement

Develop organizational awareness of AI's potential and risks, establish interim use guidelines to protect against ungoverned experimentation, and build the leadership support needed to commit to a formal readiness assessment.

Level 2 | Readiness and Planning

Building the organizational foundation required to implement AI responsibly

Overview

At Level 2, the agency has moved from curiosity to structured preparation. Leadership has committed to AI modernization as an organizational priority, and formal planning is underway across data infrastructure, workforce, governance, and funding. The agency has completed or is actively completing a readiness assessment and is using results to make realistic decisions about where to invest before piloting any AI tools.

Level 2 is where most implementation failures are either prevented or set in motion. Agencies that skip genuine readiness assessment and move directly to piloting consistently discover mid-project that their data is too fragmented, their staff lacks critical evaluation skills, or their governance structures cannot handle the decisions that arise. Investing seriously in planning at Level 2 is far less costly than recovering from a failed pilot at Level 3 or 4.

Organizational Profile at This Level

Leadership & Culture

Data Infrastructure

Workforce Capacity

Technology Systems

Governance & Legal

Senior leadership actively supports AI modernization. AI is included in strategic planning. An executive sponsor has been identified.

Readiness assessment has identified data infrastructure gaps. Key data sources have been inventoried for quality, timeliness, and accessibility.

AI literacy training plan is in development or early implementation. Staff skill gaps are documented. AI champions have been designated.

IT has begun assessing data systems for AI readiness. FedRAMP-authorized AI tools are being identified through HHS or state procurement vehicles.

AI governance structure is being designed. Interim responsible AI principles have been drafted. Legal and privacy review is underway.

Typical Activities

Completing the AI Readiness Self-Assessment (Tool 1) across six domains, with results reviewed in a cross-functional workshop.

Using readiness scores to develop a gap register with named owners and target closure dates for each identified gap.

Applying the Use Case Fit Test from the AI Project Charter (Tool 2) to screen 3 to 5 candidate use cases for feasibility and appropriate scope.

Developing a workforce training plan that defines AI literacy levels for different roles and identifies training pathways.

Drafting an initial AI roadmap (Tool 4) with sequenced phases and dependency mapping.

Mapping AI priorities to existing funding sources such as PHIG, DMI, and HRSA and identifying funding gaps.

Beginning stakeholder engagement with program leads, IT, legal, and equity staff to build shared understanding of AI opportunities and constraints.

Common Challenges at This Level

Readiness assessment results revealing more gaps than expected, creating pressure to skip foundational work and pilot prematurely.

Grant deadlines creating urgency that conflicts with deliberate planning, pushing agencies to commit to AI deliverables before they are ready.

Difficulty prioritizing which use cases to advance when many proposals exist and no decision framework has been established.

Workforce development competing with operational demands: staff who need AI literacy training are often those with the least available time.

Legal and privacy review taking longer than anticipated, especially for agencies without prior experience with AI-specific agreements.

Plays and Milestones

Relevant Plays

Key Milestones for Advancement

• Play 2: Establish AI Governance

• Play 4: Readiness Assessment

• Play 5: Workforce Development

• Play 6: Prioritize Use Cases

• Play 7: Design the Roadmap

• Play 8: Develop a Funding Strategy

• AI Readiness Self-Assessment completed with domain scores documented and shared with leadership.

• Gap register created with owners and timelines for the 5 to 8 highest-priority gaps.

• At least one blocking gap, such as data access or governance, has an active remediation plan.

• At least one candidate use case has been screened using the Use Case Fit Test and identified as a strong pilot candidate.

• Draft AI roadmap completed covering at least 12 months with sequenced activities and dependency mapping.

• At least one funding source has been aligned to AI activities in a work plan or grant application.

• An interim AI governance structure is operating with a regular meeting cadence.

Warning Sign: Stuck at This Level

The agency is stuck at Level 2 if it has been in planning mode for more than 12 to 18 months without launching a pilot. Planning that never produces action is a form of organizational avoidance, not responsible preparation.

Goal for Advancement

Complete an honest readiness assessment, address the most critical blocking gaps, identify 1 to 3 prioritized use cases with strong organizational support, and establish the governance, funding, and workforce foundations required to launch a responsible pilot.

Level 3 | Governance and Pilot Development

Establishing formal governance and running well-structured initial pilots

Overview

At Level 3, the agency is actively building governance structures and developing initial AI pilots. This is the transition from planning to doing: the agency has completed its readiness foundation and is now running time-bounded, carefully scoped pilots with defined success criteria, equity safeguards, and human oversight built in from the start.

Level 3 is where governance and implementation become mutually reinforcing or break down. Pilots without governance produce tools that cannot be scaled responsibly. Governance without pilots produces policies disconnected from operational reality. The most effective Level 3 agencies run both tracks simultaneously.

Organizational Profile at This Level

Leadership & Culture

Data Infrastructure

Workforce Capacity

Technology Systems

Governance & Legal

Executive sponsor is actively engaged. Leadership receives regular pilot progress briefings. AI governance committee is established with authority to approve, pause, or stop projects.

Pilot-specific data access agreements are in place. Data quality has been validated for the pilot use case. Data governance policies cover AI-related use.

Staff directly involved in pilots have completed role-appropriate AI literacy training. Subject matter experts are designated as required reviewers of all AI outputs.

FedRAMP-authorized AI tools are procured or in procurement. Pilot environments are separated from production systems. Data pipelines for pilot use cases are operational.

AI governance committee is operating with defined authorities. Responsible AI policies are documented and distributed. Equity impact assessments have been completed for each pilot.

Typical Activities

Completing AI Project Charters (Tool 2) for each pilot, covering problem statement, scope limits, success criteria, data requirements, equity safeguards, and go/no-go criteria.

Completing the Equity Impact Assessment Checklist (Tool 3) to identify affected populations, data bias risks, and required safeguards before any system goes live.

Building and testing pilot AI systems in controlled environments with synthetic or de-identified data before connecting to production sources.

Validating AI outputs with subject matter experts using structured review protocols, tracking error rates and hallucination incidence.

Establishing performance baselines so that pilot results can be compared to pre-AI workflows.

Conducting structured stakeholder engagement sessions to collect community input on AI system design before the pilot launches.

Common Challenges at This Level

Governance processes that are too slow for the pace of technical development, creating pressure to proceed without required approvals.

Pilot scope creep: a well-intentioned pilot expands beyond its original boundaries before governance and staffing structures can keep pace.

Equity impact assessments identifying significant risks that require design changes, creating tension with funder timeline commitments.

Subject matter expert review bottlenecks: designated reviewers lack protected time for AI output review responsibilities.

Human oversight eroding in practice: staff begin accepting AI outputs without critical evaluation after repeated early positive experiences.

Plays and Milestones

Relevant Plays

Key Milestones for Advancement

• Play 9: Change Management

• Play 10: Build and Deploy AI Solutions

• At least one AI pilot is running with a completed Project Charter, Equity Impact Assessment, and documented go/no-go criteria.

• AI governance committee has conducted at least one formal review of a pilot project with findings documented.

• A responsible AI policy covering at least the pilot use case is documented, distributed, and acknowledged by all staff involved.

• Performance baselines have been established and are being tracked for each pilot.

• An interim pilot evaluation report has been produced and reviewed by leadership and the governance committee.

Warning Sign: Stuck at This Level

The agency is stuck at Level 3 if pilots have been running for more than 18 months without a formal evaluation and go/no-go decision, or if governance structures were established on paper but are not being used to make actual decisions. Governance that does not influence practice is compliance theater, not responsible AI.

Goal for Advancement

Complete at least one time-bounded pilot with rigorous governance, equity assessment, and subject matter expert oversight. Use the evaluation results to make a defensible go/no-go decision about scaling, and use lessons learned to strengthen governance and workforce practices for the next phase.

Level 4 | Operational Deployment

Integrating validated AI capabilities into routine public health workflows

Overview

At Level 4, the agency has completed at least one pilot evaluation with a go-forward decision and is deploying AI-enabled capabilities into operational workflows. AI tools are no longer experimental; they are used by program staff as part of regular work, with governance oversight, ongoing performance monitoring, and equity auditing embedded in operational processes.

Operational deployment is where AI delivers its intended public health value, but it is also where the risks of automation bias, model drift, and equity disparities become most consequential. Tools that worked well in a controlled pilot may behave differently at scale, with different populations, or when used by staff who were not involved in the original design.

Organizational Profile at This Level

Leadership & Culture

Data Infrastructure

Workforce Capacity

Technology Systems

Governance & Legal

Leadership regularly reviews AI performance reports as part of governance oversight. AI is included in agency strategic plans and annual reports as an established capability.

Production-grade data pipelines support AI operations with documented quality standards and automated monitoring. Data governance agreements cover all AI use cases.

All staff who interact with AI outputs have completed critical evaluation training. Designated reviewers have protected time for AI output review. AI champions support peer learning.

AI tools are integrated into production data systems with appropriate security, backup, and access controls. FedRAMP-authorized platforms are in use for all production deployments handling sensitive data.

Governance committee conducts quarterly performance and equity reviews. Disclosure practices are documented and audited. Incident response protocols have been tested.

Typical Activities

Using the Performance Monitoring Dashboard (Tool 6) to track accuracy, user adoption, and equity metrics for each deployed system monthly or quarterly.

Conducting quarterly hallucination and fabrication audits for any generative AI system used in public-facing or consequential internal products.

Completing an annual equity audit for each deployed system, comparing performance across demographic subgroups and taking corrective action when disparities are found.

Scaling validated pilots to additional programs, geographies, or user groups with updated data governance agreements and staff training for each expansion.

Applying the Incident Response Checklist (Tool 7) when a deployed system produces erroneous, biased, or unexpected outputs.

Contributing lessons learned to peer networks including CSTE, ASTHO, NACCHO, and CDC's STLT Data Connection.

Common Challenges at This Level

Automation bias increasing as staff become accustomed to AI outputs: quality review processes that were rigorous at launch may relax over time without active management.

Model drift: AI systems trained on historical data may underperform as underlying populations, surveillance patterns, or data quality evolve.

Equity disparities emerging at scale that were not detectable in smaller pilots, requiring design changes or operational pauses.

Sustainment funding gaps: AI systems funded through grant pilots may face resource cliffs when grant periods end.

Staff turnover in key AI roles reducing institutional knowledge and quality oversight capacity.

Plays and Milestones

Relevant Plays

Key Milestones for Advancement

• Play 10: Build and Deploy AI Solutions

• Play 11: Monitor and Evaluate AI Systems

• At least one AI system is fully deployed in a production environment with active governance oversight.

• Performance Monitoring Dashboard data is reviewed monthly by the operational owner and quarterly by the governance committee.

• An annual equity audit has been completed for at least one deployed system, with findings documented and addressed.

• Sustainment funding for deployed systems is secured beyond the initial grant period.

• Disclosure practices for AI-assisted products are documented, trained, and audited at least annually.

Warning Sign: Stuck at This Level

The agency is stuck at Level 4 if performance monitoring has become a compliance exercise that does not influence decisions, or if equity audits are completed but disparities are not acted upon. Deployment without continuous improvement is technical debt accumulating quietly.

Goal for Advancement

Establish AI deployment as a sustainable, well-governed operational capability with documented performance, active equity monitoring, and reliable sustainment funding. Prepare the organization to expand AI use responsibly to new program areas.

Level 5 | Continuous Improvement and Innovation

Sustaining AI as a core organizational capability that advances public health

Overview

At Level 5, AI is embedded as a core organizational capability rather than a discrete project or initiative. Multiple AI-enabled tools are operating across programs, supported by mature governance, skilled staff, strong data infrastructure, and a culture of continuous improvement.

Reaching Level 5 does not mean AI work is complete or that governance can be relaxed. It means the organization has developed the capacity to work with AI responsibly, evaluate its impact honestly, and adapt quickly when problems arise. The primary risk at Level 5 is complacency: assuming that past success guarantees future safety.

Organizational Profile at This Level

Leadership & Culture

Data Infrastructure

Workforce Capacity

Technology Systems

Governance & Legal

Leadership champions responsible AI as an organizational value. AI is integrated into the agency's public health strategy, annual reporting, and budget planning. The agency contributes to national AI learning communities.

Data infrastructure is mature, interoperable, and regularly audited for quality. Automated quality monitoring supports AI system reliability. Synthetic data capabilities are available for model development without PHI exposure.

AI literacy is integrated into onboarding and continuing education for all relevant roles. Data scientists or AI-capable staff are embedded in or accessible to program teams. Communities of practice connect to national networks.

Multiple AI systems operate in production across program areas. The agency maintains internal capacity to retrain and evaluate AI systems. An AI use case inventory is maintained and reviewed annually.

Governance is adaptive: policies are updated as technologies evolve and new evidence emerges. Quarterly equity audits are routine. Disclosure practices are embedded in standard product review workflows.

Typical Activities

Conducting continuous improvement cycles for each deployed AI system on a documented cadence, including retraining schedules, performance benchmarking, and equity re-evaluation.

Evaluating long-term public health outcomes attributable to AI-assisted workflows, including whether disparities have been reduced or inadvertently widened over time.

Exploring and evaluating emerging AI capabilities using the same structured governance process applied to existing systems.

Contributing case studies, evaluation findings, and governance frameworks to national peer learning communities such as CSTE, ASTHO, NACCHO, and CDC's STLT Data Connection.

Maintaining and reporting an AI use case inventory aligned with HHS AI use case inventory requirements.

Common Challenges at This Level

Overconfidence: past success with well-governed AI systems can reduce critical scrutiny of new capabilities or expansions.

Governance fatigue: committee members who have been engaged in AI oversight for years may become less rigorous in their reviews over time.

Keeping pace with technological change: governance frameworks developed for current capabilities may not be adequate for next-generation systems.

Workforce retention: staff with advanced AI skills are in high demand and may leave for private sector roles.

Community engagement erosion: as AI workflows become routine, the active stakeholder involvement that characterized earlier stages can quietly diminish.

Plays and Milestones

Relevant Plays

Key Milestones for Advancement

• Play 11: Monitor and Evaluate AI Systems

• Play 12: Execute Governance and Oversight

• An AI use case inventory is maintained, reviewed annually, and shared with appropriate oversight bodies.

• Governance policies have been reviewed and updated within the past 12 months, with changes documented and distributed.

• Long-term outcome evaluation of at least 12 months of operational data has been completed for at least one major deployed system.

• The agency has contributed at least one documented case study, evaluation finding, or governance resource to a national peer learning network.

• All deployed AI systems have been evaluated for equity within the past 12 months, with findings and actions documented.

• Sustainment funding for all deployed systems is secured in the base budget, not solely through time-limited grants.

Warning Sign: Stuck at This Level

The agency is exhibiting Level 5 risk patterns if governance reviews have become routine sign-offs rather than substantive oversight, if new AI capabilities are being deployed without equity impact assessments, or if community engagement has stopped because AI workflows feel settled. Complacency at Level 5 can produce harms that take years to detect.

Goal for Advancement

Maintain responsible, effective, and adaptive AI-enabled public health systems that advance equity, contribute to peer learning, and demonstrate sustained public health impact. Continue applying the same governance rigor to new capabilities that was applied to the first pilots.

Correspondence with the AI Readiness Self-Assessment

The AI Maturity Model and the AI Readiness Self-Assessment (Tool 1) are designed to be used together. Tool 1 provides domain-specific numeric scores across six areas. The total score out of 72 corresponds to maturity levels as follows. Note that score ranges are approximate and should be used as general guidance rather than hard thresholds.

Maturity Level

Tool 1 Score Range

Interpretation

Recommended Next Step

Level 1

Below 30 / 72

Early stage: significant foundational work needed across most domains.

Begin Play 1 (Vision and Guardrails); establish interim acceptable-use policy.

Level 2

30 to 44 / 72

Moderate readiness: planning capacity exists but key gaps remain.

Complete readiness assessment; begin Play 3 and gap remediation.

Level 3

45 to 59 / 72

Good readiness: building blocks in place; governance and pilots underway.

Launch first pilot with strong governance; apply Plays 9 and 10.

Level 4

60 to 68 / 72

Strong readiness: operational deployment with active monitoring.

Scale successful pilots; apply Plays 11 and 12 rigorously.

Level 5

Above 68 / 72

Excellent readiness: continuous improvement and multi-program integration.

Sustain governance rigor; contribute to peer learning; evaluate emerging capabilities.

5. The 12-Play AI Implementation Framework

This section reframes AI implementation as a set of plays that STLT health departments can run. Each play describes what to do, who should be involved, and what success looks like. The tools and checklists in the companion Toolkit are the instruments you use while running each play.

The 12 plays are organized into four phases. Governance (Play 2) is established immediately after vision is set — before use cases are selected, before stakeholders are formally engaged, and before any pilot begins. This reflects a core principle: responsible AI cannot be governed after the fact. A governance structure, responsible AI policy, and oversight committee must exist to authorize every subsequent decision in the framework. Funding strategy (Play 8) is positioned after roadmap design because a funding strategy can only be built once you know what you need to fund, and must be secured before implementation begins.

PLAN PHASE — Plays 1–6: Establish vision, governance, and organizational foundation

Play 1 — Vision & Guardrails

Goal: Establish a clear strategic vision for how AI will support the agency's public health mission while defining principles that ensure responsible, ethical, and legally compliant use.

Why This Play Matters

Every AI initiative begins with leadership alignment. Without a shared vision, individual programs may pursue disconnected pilots that compete for resources, create confusion among staff, or undermine public trust. Setting guardrails early — especially around equity, human oversight, and prohibited uses — prevents downstream harms and builds institutional trust.

Key Actions

Convene a 60–90 minute leadership session to align on the agency's top 3–5 strategic problems AI should help address.

Agree on non-negotiable guardrails: human oversight requirements, equity commitments, privacy protections, and prohibited uses.

Identify conditions that require additional caution before proceeding, including when no clear governance structure exists or when legal concerns remain unresolved.

Draft a short AI Vision and Principles document (1–2 pages) including mission statement, priority problem areas, governing principles, and explicit prohibitions.

Align the vision document with the NIST AI Risk Management Framework (AI RMF 1.0) to strengthen credibility with federal funders.

Identify executive sponsorship and leadership oversight for AI initiatives.

Circulate the draft to all program staff and external partners; collect feedback; update annually.

Primary Output

AI vision and principles document

Supporting Tools

Tools 1, 3, 12, 13

Play 2 — Establish AI Governance

Goal: Create the governance structures, policies, and oversight mechanisms that ensure all AI work is authorized, accountable, and compliant — before any project begins.

Why This Play Matters

Governance is not something you add once AI is already running. It must exist before the first use case is selected, before the first vendor is engaged, and before the first pilot is approved. A governance committee without authority is a rubber stamp; a governance policy without a committee to enforce it is a document. Both must be in place — with real authority — from the start of the program.

Key Actions

Establish an AI Governance Committee (or designate an existing body) with cross-functional membership, clear authority, and a regular meeting cadence. The committee must have authority to approve new use case proposals, review equity impact assessments, authorize emergency suspension of any AI system that causes harm, and recommend policy updates.

Adopt a Responsible AI Policy using the Responsible AI Policy Template (Tool 13) that establishes the agency's binding commitments on transparency, privacy, equity, human oversight, and legal compliance.

Formally charter the committee using the AI Advisory Committee Charter (Tool 15) or AI Governance Board Charter (Tool 22), defining membership, voting authority, meeting frequency, and decision-making processes.

Align AI governance with existing data and IT governance processes to avoid duplication and ensure coordination across the agency.

Embed disclosure and scientific integrity requirements: the agency must disclose when AI is used in public-facing or significant internal products; generative AI outputs must be subject to hallucination and fabrication audits.

Establish data use agreement requirements using the Data Use Agreement Template (Tool 24) as the standard for all AI-related data arrangements with vendors and partners.

Set model validation and documentation standards using the Model Validation and Documentation Standards (Tool 23) as the governing requirement: no AI system may be deployed without completing this process.

Primary Output

Governance committee, responsible AI policy, and governance charter

Supporting Tools

Tools 3, 13, 15, 22, 23, 24

Play 3 — Engage Stakeholders

Goal: Establish a structured stakeholder engagement framework that enables internal and external partners to provide meaningful input throughout planning, implementation, and evaluation.

Why This Play Matters

Stakeholder and community engagement is not a checkbox at the end of project design; it is a continuous process that begins before use case selection and continues through deployment and monitoring. AI systems built without genuine community input often fail in deployment, even when they perform well technically.

Key Actions

Identify key stakeholder groups using the Stakeholder Mapping and RACI Template (Tool 14) and assign a staff lead responsible for each group's engagement.

For any use case with direct community impact, form a small advisory group (8–12 people) that includes community-based organizations, people with lived experience, and advocates from populations most likely to be affected.

Share clear, plain-language explanations of what the AI system will do and will not do — including how data will be protected, who will have access, and how errors will be handled.

Conduct at least one formal feedback session with community advisors before finalizing the design of any AI system that uses community data or will directly affect community members.

For any AI use case involving research about or with Tribal communities, consult with Tribal partners and follow Indigenous data sovereignty principles before proceeding.

Capture all feedback and document changes made in response to community input; share a 'what we heard and what we changed' summary back to advisors.

Compensate community advisors for their time; make engagement activities accessible with translation, flexible meeting formats, and childcare if needed.

Primary Output

Stakeholder map and engagement plan

Supporting Tools

Tools 11, 14, 15

Play 4 — Readiness Assessment

Goal: Assess the organization's current state across five domains — leadership, data infrastructure, workforce, technology, and governance — so planning is grounded in organizational reality.

Why This Play Matters

Departments that skip readiness assessment frequently discover mid-project that their data is too fragmented, their staff lacks necessary skills, or their governance structures are insufficient to support responsible AI use. A frank readiness assessment done early is far less costly than a failed implementation done later.

Key Actions

Schedule a half-day workshop with representatives from leadership, IT, data, programs, legal/privacy, equity, and community engagement. Assign a neutral facilitator to encourage candid participation.

Walk through the AI Readiness Self-Assessment (Tool 1), rating each indicator. Encourage honest scoring; this is not an audit.

Identify 5–8 high-priority gaps that must be addressed before or alongside AI pilots. Distinguish between blocking gaps (e.g., no data access) and parallel gaps (e.g., training that can occur simultaneously with a pilot).

Assign owners and timelines for closing each gap. Document as an action register reviewed monthly by the governance committee.

Confirm access to FedRAMP-authorized AI tools or identify an active procurement pathway through HHS or state procurement vehicles.

Use readiness findings to brief the AI Governance Committee on organizational constraints that will shape which use cases are feasible and on what timeline.

Primary Output

Readiness scores and gap register

Supporting Tools

Tools 1, 2, 9

Play 5 — Workforce Development

Goal: Develop the workforce capabilities needed to responsibly adopt generative and agentic AI across public health programs.

Why This Play Matters

Formal training programs alone are insufficient for sustained AI adoption. Peer learning networks and communities of practice are among the most effective mechanisms for building staff confidence, surfacing use cases, and solving problems quickly. Departments that establish internal communities of practice consistently achieve higher adoption rates than those relying on structured training alone.

Key Actions

Define AI literacy levels appropriate for different workforce roles (Aware, Proficient, Practitioner) using the AI Workforce Competency Framework (Tool 16).

Conduct a Training Needs Assessment (Tool 18) to identify gaps between current and required capabilities before designing or procuring training programs.

Designate AI champions across divisions using the AI Champion Designation and Role Guide (Tool 17).

Develop a workforce training plan that defines training pathways for each literacy level and assigns responsibility for delivery.

Establish communities of practice to support ongoing learning, including monthly AI office hours, a shared prompt library, and connections to external networks such as CSTE, ASTHO, NACCHO, and CDC's STLT Data Connection.

Primary Output

AI workforce training plan

Supporting Tools

Tools 5, 16, 17, 18

Play 6 — Prioritize Use Cases

Goal: Identify and prioritize AI use cases that align with public health priorities and organizational readiness, selecting the small number most likely to deliver meaningful impact — with governance committee approval before any work begins.

Why This Play Matters

Without a disciplined prioritization process, resources get spread across too many projects, none of which succeed. Governance committee approval before work begins ensures every project has been reviewed for equity risk, data feasibility, legal compliance, and alignment with the agency's AI vision — not just technical potential.

Key Actions

Ask each program area to propose 2–3 potential AI use cases using a simple template: problem statement, proposed AI approach, required data sources, potential benefits, and known risks.

Evaluate each candidate on consistent criteria: impact, feasibility, health equity, scalability, sustainability, stakeholder support, data availability, and regulatory clarity.

Apply the Use Case Fit Test before committing to any use case: for generative AI, confirm the task is content-driven and output will be reviewed by a subject matter expert; for agentic AI, confirm scope is clearly defined and relies only on non-sensitive data.

Use the Use Case Scoring and Prioritization Matrix (Tool 19) to score and rank candidates consistently.

Select no more than 1–3 initial use cases for pilot work. Document why each was selected and why others were deferred.

Submit selected use cases to the AI Governance Committee for formal approval before any development work begins.

Record deferred use cases in a future backlog with notes on what would need to change for them to be prioritized.

Primary Output

Governance-approved AI use case portfolio

Supporting Tools

Tools 2, 3, 11, 19

BUILD PHASE — Plays 7–9: Design the roadmap, secure resources, and manage change

Play 7 — Design the Roadmap

Goal: Translate vision, governance, readiness, and approved use case priorities into a sequenced, multi-year plan that coordinates initiatives, dependencies, and resource investments.

Why This Play Matters

Without a roadmap, AI initiatives develop as isolated silos with no coordination on shared infrastructure, governance, or workforce needs. Every roadmap milestone should reference its governance touchpoint — the committee review or policy requirement that must be satisfied before the next phase begins.

Key Actions

Complete an AI Project Charter (Tool 2) for each governance-approved use case, covering problem statement, scope, success criteria, data requirements, equity safeguards, stakeholder roles, timeline, budget, and risk register.

Lay projects onto the AI Implementation Timeline (Tool 4), adjusting durations based on readiness assessment results and use case complexity.

Identify cross-cutting dependencies explicitly — shared cloud environments, governance committee reviews, staff hiring, FedRAMP-compliant tool procurement — and show these as early milestones before pilot phases are scheduled.

Present the draft roadmap to the AI Governance Committee for validation and approval before it is finalized.

Publish a one-page roadmap summary that all staff and partners can see; maintain this as a living document updated at least every six months.

Primary Output

Governance-approved AI roadmap and implementation timeline

Supporting Tools

Tools 2, 4, 9, 10

Play 8 — Develop a Funding Strategy

Goal: Build a coordinated, multi-year funding strategy that maps each roadmap initiative to specific funding sources, staffing plans, and infrastructure investments — ensuring the roadmap is operationally grounded, not aspirational fiction.

Why This Play Matters

The most common cause of AI project failure in public health is resource misalignment: projects are approved without dedicated staff, funded for one phase but not the next, or dependent on infrastructure that was never actually built. This play follows roadmap design because a funding strategy can only be built once you know what you need — and must be secured before implementation begins. See Section 3 of this playbook for detailed guidance on CDC DMI, PHIG, HRSA, NIH/NSF, and other funding streams that can be braided to support AI work.

Key Actions

Map each AI project and cross-cutting activity to specific funding sources using the Funding Strategy Checklists (Tool 8). Create a simple funding map: list each project and resource category, then show which funding source covers each cell.

For each project, complete a resource requirements table listing required FTE by role, contractor needs, infrastructure costs, and evaluation budget by project phase.

Update DMI, PHIG, and other grant work plans to explicitly include AI activities, budget lines, and deliverables that mirror the AI roadmap.

Ensure every AI project has a confirmed funding source for each phase: pilot, scale, and sustainment — with no phase dependent on funding that has not been secured.

Build multi-year sustainment costs — including hosting, model retraining, staff monitoring time, and ongoing equity audits — into base operating budgets rather than relying solely on time-limited grant funding.

Develop contingency plans for each project: what happens if a key staff member leaves, if a funding source is reduced, or if infrastructure timelines slip?

Review resource allocation quarterly and adjust as AI projects encounter unexpected challenges or technology changes.

Primary Output

Funding portfolio and grant alignment

Supporting Tools

Tools 4, 8, 10

Play 9 — Change Management

Goal: Prepare the organization for adoption of AI-enabled tools and workflows by addressing workforce readiness, organizational culture, and staff concerns before and during deployment.

Why This Play Matters

The most common cause of AI project failure in public health is not technical — it is organizational. Staff resistance, inadequate training, and unclear communication undermine even technically sound systems. Change management planning should happen before implementation begins, not after adoption struggles emerge.

Key Actions

Communicate AI goals and anticipated benefits clearly to staff and leadership using the Staff Communication and FAQ Template (Tool 20).

Develop a structured Change Management Plan (Tool 5) that identifies training needs, workflow changes, and adoption support for each user group affected by an AI project.

Use the Resistance and Barrier Identification Tool (Tool 21) to proactively surface staff and organizational concerns before pilot launch.

Develop strategies to support workforce engagement and acceptance, including change champions with protected time to support colleagues.

Address operational or cultural barriers to implementation identified through the barrier assessment.

Establish feedback mechanisms so staff can raise concerns, report errors, and suggest improvements throughout the pilot.

Coordinate change management planning with governance committee: staff concerns and adoption barriers should be tracked and reported back to oversight.

Primary Output

Adoption strategy and communications plan

Supporting Tools

Tools 5, 20, 21

IMPLEMENT PHASE — Play 10: Build, pilot, deploy, and scale AI solutions

Play 10 — Build and Deploy AI Solutions

Goal: Develop, test, pilot, deploy, and scale AI solutions that support governance-approved use cases while adhering to all governance policies, equity requirements, and validation standards.

Why This Play Matters

The pilot phase is about learning, not proving. Keep pilots small, time-bounded, and deliberately limited in scope to minimize risk and accelerate learning. Every transition — from pilot to scale, and from scale to sustainment — requires explicit governance committee approval. Scaling is not automatic; it is an authorized decision.

Key Actions

Establish technical infrastructure and data pipelines required for AI development, using the Technical Infrastructure Readiness Checklist (Tool 25) before development begins.

Build and test pilot AI systems in controlled environments with synthetic or de-identified data before connecting to production sources; use the AI Model Testing and Validation Protocol (Tool 26) and complete the full Model Validation and Documentation Standards (Tool 23) before any deployment.

Run pilots for no more than 90–120 days before a formal evaluation using the AI Pilot Evaluation Dashboard (Tool 27) and make a go/no-go decision using the Pilot-to-Scale Decision Framework (Tool 28).

Document everything during the pilot: what worked, what failed, what was surprising, and what would need to change to scale.

Before scaling, complete a formal equity impact assessment to ensure pilot results hold across different populations.

Submit pilot evaluation results and scale-up plans to the AI Governance Committee for approval before expanding deployment.

Deploy solutions into operational public health workflows with updated data governance agreements, privacy impact assessments, and vendor contracts reflecting the expanded scope.

Primary Output

AI pilot and production deployments

Supporting Tools

Tools 2, 4, 7, 9, 10, 23, 24, 25, 26, 27, 28

SUSTAIN PHASE — Plays 11–12: Monitor, evaluate, and continuously govern AI systems

Play 11 — Monitor and Evaluate AI Systems

Goal: Assess whether deployed AI systems perform as expected and deliver measurable public health value, with continuous attention to equity and ongoing governance reporting.

Why This Play Matters

Deployment without monitoring is how AI systems cause harm quietly. Models trained on historical data can drift as underlying populations change. Governance protocols must actively counteract automation bias — the tendency of staff who work alongside AI systems to reduce their critical scrutiny over time. Monitoring is not an administrative task; it is a core governance function.

Key Actions

Monitor model performance and reliability on a continuous basis using the Performance Monitoring Dashboard (Tool 6) — reviewed monthly by the operational owner and quarterly by the governance committee.

Track equity metrics continuously using the Equity and Disparity Impact Monitoring Tool (Tool 29) — monitoring subgroup performance by race/ethnicity, geography, age, and language.

Conduct quarterly hallucination and fabrication audits for any generative AI system used in public-facing or consequential internal products.

Evaluate public health outcomes influenced by AI insights using the AI Evaluation and Outcomes Reporting Template (Tool 30).

Conduct a formal annual equity audit for each deployed system, comparing performance across demographic subgroups and taking corrective action when disparities are found.

When disparate impact is detected, escalate immediately to the governance committee and document the investigation and response.

Refine models and workflows based on evaluation findings; use the Continuous Improvement Log (Tool 31) to capture and track all improvement actions.

Primary Output

Performance and equity monitoring reports

Supporting Tools

Tools 6, 29, 30, 31

Play 12 — Execute Governance and Oversight

Goal: Sustain responsible AI as a continuous organizational function through regular governance reviews, policy maintenance, compliance auditing, and ongoing accountability.

Why This Play Matters

Governance does not end when an AI system goes live — it intensifies. The governance committee's role in the sustain phase is to ensure that accountability is maintained, that equity monitoring findings drive real changes, that policies stay current with evolving technology, and that the organization can honestly answer the question: is this AI system still doing what we said it would do, for whom we said it would do it?

Key Actions

Conduct AI Governance Committee reviews on a regular cadence using the AI Governance Review Meeting Template (Tool 32) — monthly during active pilot phases; quarterly once systems are in sustainment.

Ensure compliance with responsible AI policies and legal requirements through annual compliance audits using the Compliance Audit Checklist (Tool 33).

Maintain documentation, validation records, and audit trails; use the Policy Update and Version Control Log (Tool 34) to track all AI governance policy changes.

Update governance policies as technologies and organizational needs evolve — conduct annual reviews against evolving federal guidance including CDC STLT resources, NIST AI RMF updates, and relevant HHS directives.

Define sunset criteria for each AI system: under what conditions would it be retired or replaced?

Assign a named operational owner responsible for ongoing performance and governance compliance for each deployed system.

Use evaluation evidence to support sustained AI investment in legislative and budget processes, translating technical performance metrics into public health impact terms.

Primary Output

Audits, compliance records, and governance documentation

Supporting Tools

Tools 6, 7, 31, 32, 33, 34

6. Case Studies and Examples

The following case studies illustrate what it looks like when the plays and tools in this playbook are put into practice. They are illustrative examples drawn from real-world public health AI implementations.

Case Study 1: Outbreak Early Warning System

A state health department used an LLM-assisted agentic AI system to enhance early outbreak detection by automatically processing incoming syndromic surveillance data, laboratory reports, and emergency department chief complaints, flagging geographic clusters for human epidemiologist review. The department began with Play 1 (Vision and Guardrails) to establish that the system would only flag geographic clusters for human investigation — never trigger automatic notifications to the public or enforcement actions. Play 3 (Readiness Assessment) revealed that the department's syndromic surveillance data had a 48-hour reporting lag, which was addressed through a DMI-funded data pipeline investment before the pilot began.

Results: The system detected three potential outbreak clusters in its first year that were confirmed on investigation. Average detection time decreased by 4 days compared to the previous manual review process. Community engagement (Play 2) surfaced concerns from immigrant communities about data use, which led to explicit language in the department's AI principles prohibiting sharing of individual-level data with law enforcement.

Case Study 2: Vaccine Forecasting and Communication

A territorial health department used generative AI to improve vaccine demand communication and planning by automatically drafting tailored outreach messages for different geographic areas, age groups, and provider types. The project was funded through a braided combination of PHIG (for the data analytics environment) and immunization cooperative agreement funds (for the specific forecasting application) — reflecting the funding strategy approach described in Play 8. Play 5 (Use Case Prioritization) was critical: the initial use case inventory included 11 potential AI applications, and the prioritization process identified vaccine forecasting as the strongest candidate because of its clear ROI, high stakeholder support, and available data.

Results: The department significantly improved the speed and quality of vaccine communications, reducing draft turnaround time by 65% and improving fill rates at high-risk community sites. The project charter (Play 7 tool) was used to negotiate clear success criteria with leadership before the pilot began, which provided a defensible basis for the go-forward decision at the six-month review.

Case Study 3: Environmental Health Monitoring

A large local health department used generative AI tools to analyze housing inspection reports and automatically identify patterns in free-text notes that flagged potential lead exposure risks in older housing stock. Play 2 (Stakeholder and Community Engagement) was particularly important for this use case: communities with older housing stock are often communities of color with historical reasons to distrust government data collection. The department formed a community advisory board that included tenant rights advocates, housing advocates, and residents from the highest-risk census tracts.

Results: The advisory board identified privacy concerns about the potential use of imagery to identify unauthorized units, which led to a formal prohibition on sharing AI outputs with housing code enforcement. This governance decision, embedded in the project charter through Play 7, prevented a significant breach of community trust before it occurred.

Case Study 4: Chronic Disease Risk Prediction in a Tribal Context

A tribal health department used an agentic AI system to identify tribal members at elevated risk for type 2 diabetes complications and automate the outreach workflow for care management — while keeping all data under tribal jurisdiction and all final decisions with tribal health staff. Play 1 (Vision and Guardrails) was especially critical in this tribal sovereignty context: the department's AI principles explicitly stated that all data remained under tribal jurisdiction, that no data would be shared with state or federal systems without explicit tribal council approval, and that AI tools would be operated exclusively by tribal health staff.

Results: Care management outreach rates increased by 35% among high-risk members. The governance framework was adapted to incorporate tribal council oversight as the primary governance body, with a reporting relationship to the tribal health board rather than the state health department. This model is now being shared with other tribal health departments as a replicable governance design.

Case Study 5: Health Communications in a Mid-Size City

A mid-size city health department implemented a secure, internal generative AI tool to draft plain-language health content based on prompts and approved reference materials. The system was configured to generate messages at specific reading levels and in multiple languages, use department-approved style and terminology, and flag content that might include unverified claims for extra review. The Use Case Prioritization play helped leadership select this as an early generative AI project because it had clear benefits, limited technical risk, and strong staff demand.

Results within six months: average turnaround time for draft materials fell from 10 business days to 3; staff reported spending more time on strategy and partner relationships; materials were consistently produced at the targeted reading level and in multiple languages; and the department was able to respond more quickly to emerging issues with timely, consistent messaging. Key lesson: generative AI is most effective when anchored in clearly defined guardrails, approved source materials, and mandatory human review.

About AI Playbook for Public Health

AI Playbook for Public Health is an open resource designed to support state, territorial, local, and tribal health departments in building the leadership, workforce, and operational capacity needed to implement generative and agentic AI responsibly. This resource is updated periodically to reflect evolving technology, policy, and public health practice.